

Lecture 3

Multiple Linear Regression: Inference and Prediction

Reading: Faraway 2014 Chapters 3.1-3.2; 3.5; 4.1-4.2; 4.4;
7.3. ISLR 2021 Chapter 3.2

STAT 8020 Statistical Methods II

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Agenda

Multiple Linear
Regression:
Inference and
Prediction



General Linear F -Test

Prediction

Multicollinearity

1 General Linear F -Test

2 Prediction

3 Multicollinearity

Review: t -Test and F -Test in Linear Regression

- t -Test: Testing one predictor

- 1 **Null/Alternative Hypotheses:** $H_0 : \beta_j = 0$ vs. $H_a : \beta_j \neq 0$

- 2 **Test Statistic:** $t^* = \frac{\hat{\beta}_j - 0}{SE(\hat{\beta}_j)}$

- 3 Reject H_0 if $|t^*| > t_{1-\alpha/2, n-p}$

- Overall F -Test: Test of all the predictors

- 1 $H_0 : \beta_1 = \beta_2 = \dots = \beta_{p-1} = 0$

- 2 $H_a : \text{at least one } \beta_j \neq 0, 1 \leq j \leq p-1$

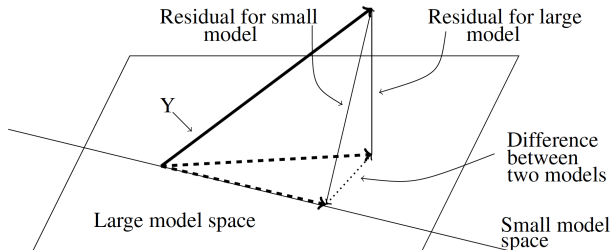
- 3 **Test Statistic:** $F^* = \frac{MSR}{MSE}$

- 4 Reject H_0 if $F^* > F_{1-\alpha, p-1, n-p}$

Both tests are special cases of **General Linear F -Test**

- Comparison of a “full model” and “reduced model” that involves **a subset of full model predictors**
- Consider a full model with k predictors and reduced model with ℓ predictors ($\ell < k$)
- Test statistic: $F^* = \frac{(SSE_{\text{reduce}} - SSE_{\text{full}})/(k - \ell)}{SSE_{\text{full}}/(n - k - 1)} \Rightarrow$ Testing H_0 that the regression coefficients for the extra variables are all zero
 - Example 1: $x_1, x_2, \dots, x_{p-1}$ vs. intercept only $\Rightarrow$ Overall F -test
 - Example 2: $x_j, 1 \leq j \leq p - 1$ vs. intercept only $\Rightarrow t$ -test for β_j
 - Example 3: x_1, x_2, x_3, x_4 vs. $x_1, x_3 \Rightarrow H_0 : \beta_2 = \beta_4 = 0$

Geometric Illustration of General Linear F -Test



Source: Faraway, *Linear Models with R*, 2014, p.34

Key idea:

- Reduced model: smaller subspace; Full model: larger subspace
- F-test measures reduction in error when adding variables

Interpretation:

- Large reduction $\Rightarrow$ added variables matter
- Small reduction $\Rightarrow$ little to no added value

Reduce Model: Species ~ Elevation

```
> summary(gala_fit1)
```

```
Call:  
lm(formula = Species ~ Elevation)
```

```
Residuals:
```

Min	1Q	Median	3Q	Max
-218.319	-30.721	-14.690	4.634	259.180

```
Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	11.33511	19.20529	0.590	0.56
Elevation	0.20079	0.03465	5.795	3.18e-06 ***

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 78.66 on 28 degrees of freedom
```

```
Multiple R-squared:  0.5454,    Adjusted R-squared:  0.5291
```

```
F-statistic: 33.59 on 1 and 28 DF,  p-value: 3.177e-06
```

In SLR: t -test $\equiv F$ -test $\Rightarrow$ identical p -values

Full model: Species ~ Elevation + Area**Question:** Does `Area` provide additional explanatory power?

```
> summary(gala_fit2)
```

Call:

```
lm(formula = Species ~ Elevation + Area)
```

Residuals:

Min	1Q	Median	3Q	Max
-192.619	-33.534	-19.199	7.541	261.514

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	17.10519	20.94211	0.817	0.42120
Elevation	0.17174	0.05317	3.230	0.00325 **
Area	0.01880	0.02594	0.725	0.47478

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 79.34 on 27 degrees of freedom

Multiple R-squared: 0.554, Adjusted R-squared: 0.521

F-statistic: 16.77 on 2 and 27 DF, p-value: 1.843e-05

Observation:

- Elevation: statistically significant
- Area: not statistically significant ($p \approx 0.47$)

Suggestion: Limited evidence that `Area` contributes beyond
`Elevation`

Performing a General Linear F -Test

● $H_0 : \beta_{\text{Area}} = 0$ vs. $H_a : \beta_{\text{Area}} \neq 0$

● $F^* = \frac{(173254 - 169947)/(2-1)}{169947/(30-2-1)} = 0.5254$

● P-value: $P[F > 0.5254] = 0.4748$, where $F \sim F_{\underbrace{1}_{k-\ell}, \underbrace{27}_{n-k-1}}$

```
> anova(gala_fit1, gala_fit2)
```

Analysis of Variance Table

Model 1: Species ~ Elevation

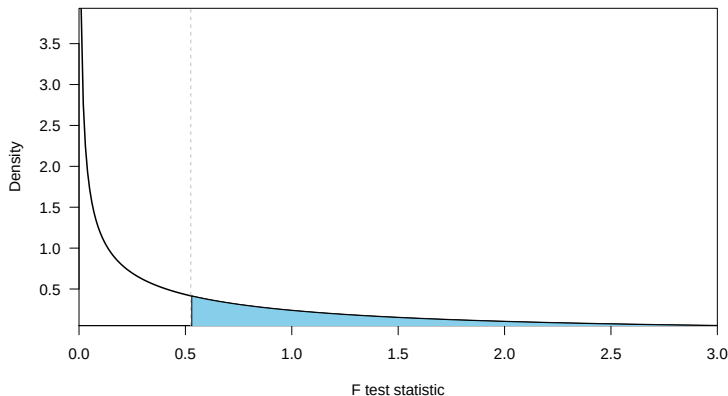
Model 2: Species ~ Elevation + Area

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	28	173254				
2	27	169947	1	3307	0.5254	0.4748

Interpretation: No significant improvement from adding Area

Note: With one additional predictor, the general linear F -test is equivalent to the individual t -test in the full model (same p -value)

Visualizing p -value



Interpretation: p -value = area to the right of F^* , measuring its extremeness under H_0

Small p -value $\Rightarrow$ strong evidence against $H_0 : \beta_{\text{Area}} = 0$

Another Example of General Linear F -Test

Full Model:

```
> full <- lm(Species ~ Area + Elevation + Nearest + Scruz + Adjacent,  
  data = gala)  
> anova(full)
```

Analysis of Variance Table

Response: Species

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Area	1	145470	145470	39.1262	1.826e-06 ***
Elevation	1	65664	65664	17.6613	0.0003155 ***
Nearest	1	29	29	0.0079	0.9300674
Scruz	1	14280	14280	3.8408	0.0617324 .
Adjacent	1	66406	66406	17.8609	0.0002971 ***
Residuals	24	89231	3718		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Reduced Model:

```
> reduced <- lm(Species ~ Elevation + Adjacent)  
> anova(reduced)
```

Analysis of Variance Table

Response: Species

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Elevation	1	207828	207828	56.112	4.662e-08 ***
Adjacent	1	73251	73251	19.777	0.0001344 ***
Residuals	27	100003	3704		

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Performing a General Linear F -Test

Goal: Do Area, Nearest, and Scruz jointly improve the model?

- Null and alternative hypotheses:

$$H_0 : \beta_{\text{Area}} = \beta_{\text{Nearest}} = \beta_{\text{Scrutz}} = 0$$

$$H_a : \text{at least one of the three coefficients} \neq 0$$

- $F^* = \frac{(100003-89231)/(5-2)}{89231/(30-5-1)} = 0.9657$

- $p\text{-value: } P[F > 0.9657] = 0.425, \text{ where } F \sim F_{3,24}$

```
> anova(reduced, full)
```

Analysis of Variance Table

Model 1: Species ~ Elevation + Adjacent

Model 2: Species ~ Area + Elevation + Nearest + Scrutz + Adjacent

	Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1	27	100003				
2	24	89231	3	10772	0.9657	0.425

Takeaway: Limited evidence that these predictors jointly improve the model

Given a new set of predictors, $\mathbf{x}_0 = (1, x_{0,1}, x_{0,2}, \dots, x_{0,p-1})^T$, the predicted response is

$$\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_{0,1} + \hat{\beta}_2 x_{0,2} + \dots + \hat{\beta}_{p-1} x_{0,p-1}.$$

Again, we can use matrix representation to simplify the notation

$$\hat{y}_0 = \mathbf{x}_0^T \hat{\boldsymbol{\beta}},$$

where $\mathbf{x}_0^T = (1, x_{0,1}, x_{0,2}, \dots, x_{0,p-1})$

We will use this formula to carry out two different kinds of predictions

Two Kinds of Predictions

For a given x_0 , there are two types of predictions:

- **Predicting the mean response**

$$E(y_0) = x_0^T \beta \quad \Rightarrow \quad \widehat{E(y_0)} = x_0^T \hat{\beta}$$

Uncertainty: from estimating β only

- **Predicting a future response**

Model: $y_0 = x_0^T \beta + \varepsilon$

$$\hat{y}_0 = x_0^T \hat{\beta}$$

Uncertainty: from estimating β and noise ε

Key idea: Same predictor, but different sources of uncertainty

Next: Compare their [prediction uncertainties](#)

Prediction Uncertainty

From Slides 2 (page 22), we have

$$\text{Var}(\hat{\beta}) = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}$$

Therefore,

$$\text{Var}(\hat{y}_0) = \text{Var}(\mathbf{x}_0^T \hat{\beta}) = \sigma^2 \mathbf{x}_0^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_0$$

We now construct $100(1 - \alpha)\%$ intervals for the two types of predictions:

- **Mean response** $E(y_0)$:

$$\mathbf{x}_0^T \hat{\beta} \pm t_{n-p, \alpha/2} \hat{\sigma} \sqrt{\mathbf{x}_0^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_0}$$

- **Future response** y_0 :

$$\mathbf{x}_0^T \hat{\beta} \pm t_{n-p, \alpha/2} \hat{\sigma} \sqrt{1 + \mathbf{x}_0^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_0}$$

The extra +1 accounts for the noise term ε

Example: Predicting Body Fat (Faraway 2014 Chapter 4.2)

```
lm(formula = brozek ~ age + weight + height + neck + chest +  
    abdom + hip + thigh + knee + ankle + biceps + forearm + wrist,  
    data = fat)
```

Residuals:

Min	1Q	Median	3Q	Max
-10.264	-2.572	-0.097	2.898	9.327

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-15.29255	16.06992	-0.952	0.34225
age	0.05679	0.02996	1.895	0.05929 .
weight	-0.08031	0.04958	-1.620	0.10660
height	-0.06460	0.08893	-0.726	0.46830
neck	-0.43754	0.21533	-2.032	0.04327 *
chest	-0.02360	0.09184	-0.257	0.79740
abdom	0.88543	0.08008	11.057	< 2e-16 ***
hip	-0.19842	0.13516	-1.468	0.14341
thigh	0.23190	0.13372	1.734	0.08418 .
knee	-0.01168	0.22414	-0.052	0.95850
ankle	0.16354	0.20514	0.797	0.42614
biceps	0.15280	0.15851	0.964	0.33605
forearm	0.43049	0.18445	2.334	0.02044 *
wrist	-1.47654	0.49552	-2.980	0.00318 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.988 on 238 degrees of freedom
Multiple R-squared: 0.749, Adjusted R-squared: 0.7353
F-statistic: 54.63 on 13 and 238 DF, p-value: < 2.2e-16

General Linear F -Test

Prediction

Multicollinearity

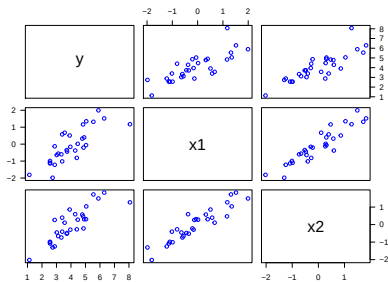
What is our prediction for the future response of a “typical” (e.g., each predictor takes its median value) man?

Predicting “Typical” Man Body Fat

- 1 Calculate the median for each predictor to get x_0
- 2 Compute the predicted value $\hat{y}_0 = x_0^T \hat{\beta}$
- 3 **CI:** uncertainty in the mean response; **PI:** uncertainty in an individual outcome $\Rightarrow$ always report prediction + interval

```
> X <- model.matrix(lmod)
> (x0 <- apply(x, 2, median))
(Intercept)    age    weight    height    neck    chest    abdom
      1.00    43.00    176.50    70.00    38.00    99.65    90.95
      hip    thigh    knee    ankle    biceps    forearm    wrist
     99.30    59.00    38.50    22.80    32.05    28.70    18.30
> (y0 <- sum(x0 * coef(lmod)))
[1] 17.49322
> predict(lmod, new = data.frame(t(x0)))
1
17.49322
> predict(lmod, new = data.frame(t(x0)), interval = "prediction")
      fit      lwr      upr
1 17.49322 9.61783 25.36861
> predict(lmod, new = data.frame(t(x0)), interval = "confidence")
      fit      lwr      upr
1 17.49322 16.94426 18.04219
```

Multicollinearity



```
> cor(sim1)
```

	y	x1	x2
y	1.0000000	0.7987777	0.8481084
x1	0.7987777	1.0000000	0.9281514
x2	0.8481084	0.9281514	1.0000000

What to notice: x_1 and x_2 are highly correlated $\Rightarrow$ hard to separate their effects

Implication: Unstable coefficients; unreliable signs and magnitudes

Multicollinearity: strong correlations among predictor variables

- **Numerical issue:** $X^T X$ is nearly singular $\Rightarrow$ inversion is unstable, leading to high variability in $\hat{\beta}$
- **Statistical consequences:**
 - Coefficients (β 's) are poorly estimated $\Rightarrow$ unstable / misleading estimates
 - R^2 and predictions may still look reasonable

A Simulated Example

Suppose the true relationship between y and predictors (x_1, x_2) is

$$y = 4 + 0.8x_1 + 0.6x_2 + \varepsilon,$$

where $\varepsilon \sim N(0, 1)$ and x_1, x_2 are **highly correlated** ($\rho = 0.9$).

Key feature: strong correlation between predictors $\Rightarrow$ potential multicollinearity

We fit the following models:

- **Model 1:**

$$y = \beta_0 + \beta_1x_1 + \beta_2x_2 + \varepsilon_1$$

True model (parameters unknown)

- **Model 2:**

$$y = \beta_0 + \beta_1x_1 + \varepsilon_2$$

Misspecified model (omits correlated predictor x_2)

Model 1 Fit

Call:

```
lm(formula = Y ~ X1 + X2)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-1.91369	-0.73658	0.05475	0.87080	1.55150

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.0710	0.1778	22.898	< 2e-16 ***
X1	2.2429	0.7187	3.121	0.00426 **
X2	-0.8339	0.7093	-1.176	0.24997

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9569 on 27 degrees of freedom

Multiple R-squared: 0.673, Adjusted R-squared: 0.6488

F-statistic: 27.78 on 2 and 27 DF, p-value: 2.798e-07

Takeaway: Even with the true model, highly correlated predictors make individual effects hard to separate, leading to unstable coefficient estimates despite a well-captured combined effect

Model 2 Fit

Call:

```
lm(formula = Y ~ X1)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-2.09663	-0.67031	-0.07229	0.87881	1.49739

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	4.0347	0.1763	22.888	< 2e-16 ***
X1	1.4293	0.1955	7.311	5.84e-08 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.9634 on 28 degrees of freedom

Multiple R-squared: 0.6562, Adjusted R-squared: 0.644

F-statistic: 53.45 on 1 and 28 DF, p-value: 5.839e-08

Takeaway: Omitting correlated $x_2 \Rightarrow$ its effect is absorbed by x_1 , leading to a biased (aggregated) coefficient

Takeaways

Model 1 fit:

```
Call:
lm(formula = Y ~ X1 + X2)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-1.91369 -0.73658  0.05475  0.87080  1.55150
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  4.0710     0.1778   22.898 < 2e-16 ***
X1           2.2429     0.7187    3.121  0.00426 **
X2          -0.8339     0.7093   -1.176  0.24997
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.9569 on 27 degrees of freedom
Multiple R-squared:  0.673,    Adjusted R-squared:  0.6488
F-statistic: 27.78 on 2 and 27 DF,  p-value: 2.798e-07
```

Recall the true model:

$$y = 4 + 0.8x_1 + 0.6x_2 + \varepsilon,$$

where $\varepsilon \sim N(0, 1)$, x_1 and x_2 are positively correlated with $\rho = 0.9$

Model 2 fit:

```
Call:
lm(formula = Y ~ X1)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-2.09663 -0.67031 -0.07229  0.87881  1.49739
```

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  4.0347     0.1763   22.888 < 2e-16 ***
X1           1.4293     0.1955    7.311 5.84e-08 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
Residual standard error: 0.9634 on 28 degrees of freedom
Multiple R-squared:  0.6562,    Adjusted R-squared:  0.644
F-statistic: 53.45 on 1 and 28 DF,  p-value: 5.839e-08
```

Summary:

- β 's are not well estimated in model 1
- Spurious regression coefficient estimates
- In model 2, R^2 and predicted values are OK compared to model 1

Variance Inflation Factor (VIF)

We use the variance inflation factor (VIF)

$$\text{VIF}_i = \frac{1}{1 - R_i^2}$$

to quantify multicollinearity in MLR, where R_i^2 is from regressing X_i on the other predictors

Interpretation:

- $\sqrt{\text{VIF}_i}$: factor by which the standard error of $\hat{\beta}_i$ is inflated

Rule of thumb:

- $\text{VIF} \approx 1$: no multicollinearity
- $\text{VIF} > 5$: moderate concern
- $\text{VIF} > 10$: serious multicollinearity

R example code

```
> library(faraway)
> vif(sim1[, 2:3])
      x1      x2
7.218394 7.218394
```

Summary

These slides cover:

- General Linear F -Test provides a unifying framework for hypothesis tests
- Making predictions and quantifying prediction uncertainty
- Multicollinearity and its implications for MLR

R commands:

- `anova` for model comparison based on F -test
- `predict`: obtain predicted values from a fitted model
- `vif` under the `faraway` library: computes the variance inflation factors